$$
\begin{aligned}
a_j &= \int \phi_j R\,dt - \tau \int \phi_j' R\,dt, \\
b_j &= \rho \left(\zeta \int \phi_j T\,dt - \xi\tau \int \phi_j' T\,dt \right), \\
z_j &= \int \phi_j e^{-t/\tau}\,dt, \quad \widehat{c}_0 = \frac{\sum_j z_j (a_j - b_j)}{\sum_j z_j^2}, \\
D_{\mathrm{Hb}}^{\mathrm{demo}}(\tau) &= \frac{\sum_j (a_j - b_j - \widehat{c}_0 z_j)^2}{\sum_j (\int \phi_j R\,dt)^2 + \rho^2 \sum_j (\int \phi_j T\,dt)^2}.
\end{aligned}
$$